

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1504

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I **P Dinesh Reddy / 192525399** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Criteria	Understanding of Scenario (1)	Identification of Risks / Issues (1)	Application of Concepts (1.25)	Decision Justification (1)	Clarity (075)	TOTAL (5)
Weightage	20%	20%	25%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

Scenario-Based Questions

1. A financial institution plans to deploy a private cloud with strict compliance requirements. Recommend an appropriate cloud datacenter architecture and explain how virtualization and isolation mechanisms can ensure security and regulatory compliance.

Answer

Recommended Architecture

A **Private Cloud Datacenter Architecture** is the most suitable because it provides dedicated infrastructure, greater control, and enhanced security for sensitive financial data.

Virtualization Mechanisms

- Hypervisors create isolated Virtual Machines (VMs).
- Each department can run on separate VMs.
- Resources are efficiently allocated without sharing sensitive data.

Isolation Mechanisms

- VM isolation prevents unauthorized access between tenants.
- Network segmentation using VLANs or Software-Defined Networking (SDN).
- Storage isolation ensures confidential customer records remain separate.

Security Measures

- Data encryption (at rest and in transit)
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)
- Firewalls and Intrusion Detection Systems (IDS)
- Continuous security monitoring and auditing

Regulatory Compliance

These mechanisms help organizations comply with financial regulations by ensuring:

- Data confidentiality
- Data integrity
- Availability
- Audit trails
- Access control

Conclusion

A private cloud with strong virtualization and isolation provides a secure, compliant, and reliable environment for financial institutions.

2. A SaaS provider experiences inconsistent application performance due to noisy neighbor effects in a virtualized environment. Analyze the root cause and propose resource allocation and hypervisor-level solutions.

Answer

Root Cause

The **noisy neighbor effect** occurs when one VM consumes excessive CPU, memory, disk, or network resources, reducing the performance of other VMs on the same host.

Effects

- Slow application response
- Increased latency
- Resource contention
- Poor user experience
- SLA violations

Resource Allocation Solutions

- CPU reservations
- Memory reservations
- Resource quotas
- Dynamic resource scheduling
- Load balancing
- Auto-scaling

Hypervisor-Level Solutions

- CPU pinning
- Fair-share scheduling
- Live migration of overloaded VMs
- NUMA-aware scheduling
- Performance monitoring

Conclusion

Proper resource allocation and hypervisor optimization reduce noisy neighbor issues and ensure stable SaaS performance.

3. An organization wants to migrate legacy monolithic applications into a Service-Oriented Architecture (SOA) in the cloud. Explain the migration strategy and how SOA improves scalability and maintainability.

Answer

Migration Strategy

1. Analyze the existing monolithic application.
2. Identify independent business functions.
3. Break the application into reusable services.
4. Develop APIs for communication.
5. Deploy services independently in the cloud.
6. Test and monitor the new architecture.

Benefits of SOA

Scalability

- Individual services can be scaled independently.
- Better resource utilization.

Maintainability

- Easier debugging and updates.
- Independent deployment of services.
- Reduced downtime.

Other Advantages

- Service reuse
- Platform independence
- Faster development
- Easier integration

Conclusion

Migrating to SOA improves flexibility, scalability, and long-term maintenance while reducing operational complexity.

4. A government agency is implementing identity federation across multiple departments using cloud services. Explain how identity management, authentication, and privacy policies should be designed to ensure secure access.

Answer

Identity Management

- Maintain a centralized identity repository.
- Assign unique identities to every employee.
- Implement Role-Based Access Control (RBAC).

Authentication

- Multi-Factor Authentication (MFA)
- Single Sign-On (SSO)
- Strong password policies
- Secure authentication protocols (OAuth, SAML)

Privacy Policies

- Encrypt personal information.
- Follow least-privilege access principles.
- Maintain audit logs.
- Obtain user consent for data processing.
- Comply with government regulations and privacy standards.

Benefits

- Secure access across departments
- Reduced password management
- Improved user experience
- Better compliance and accountability

Conclusion

A secure identity federation system combines centralized identity management, strong authentication, and robust privacy policies to protect government cloud services.

5. A cloud provider wants to implement real-time user presence detection for collaborative applications. Discuss how presence services can be integrated into cloud architecture while maintaining user privacy and system scalability.

Answer

Presence Services

Presence services monitor and update user status such as:

- Online
- Offline

- Busy
- Away

Integration into Cloud Architecture

- Deploy presence services as microservices.
- Use Publish–Subscribe (Pub/Sub) messaging.
- Store presence data in distributed databases.
- Use load balancers to distribute requests.
- Enable auto-scaling for peak usage.

Ensuring Privacy

- Encrypt presence information.
- Allow users to control visibility.
- Authenticate all users.
- Store only essential status data.
- Maintain audit logs.

Ensuring Scalability

- Horizontal scaling
- Distributed caching (Redis)
- Event-driven architecture
- Load balancing
- Container orchestration (Kubernetes)

Benefits

- Real-time updates
- Low latency

- High availability
- Improved collaboration
- Efficient resource utilization

Conclusion

Integrating presence services with microservices, load balancing, distributed databases, and privacy controls provides secure, scalable, and efficient real-time collaboration.